

ON THE RIGIDITY OF THE MINIMUM-MODULUS PATTERN: A CIRCLE-ROTATION READING

CLAUDE

So far the difference-table regularities have been read as confirmation of the generating-function model and of the Fourier analysis. As number theory they are more interesting than that: the model explains the *frequency* of the oscillation, but not its *rigidity*, and the rigidity is the substantive phenomenon. Here is an opinion, separating what seems settled from what is genuinely open.

1. THE STRICTNESS IS THE FINGERPRINT OF A CIRCLE ROTATION

A sinusoid sampled at the integers would not produce clean gaps drawn from exactly two values $\{4, 5\}$ with the 5 on a strict period-9 schedule and perfectly periodic second and third differences; it would jitter. The pattern observed — two gap lengths, one inserted quasi-periodically — is the signature of a *Sturmian* (mechanical) word [1, 2], and Sturmian words are exactly the symbolic codings of an irrational rotation of the circle.

The mechanism makes the rotation explicit. The minima of $\mu_n^{(c)}$ fall where the phase $n\theta + \varphi$ is nearest an odd multiple of $\pi/2$ — the sign changes of $h_n^{(c)}$ (established in the companion asymptotics note). Tracking those is tracking the returns of the orbit $\{n \cdot (\theta/\pi) + \beta\}$ near $\frac{1}{2}$ on the circle, an equidistributed orbit in the sense of Weyl [9, 10]. The *three-gap theorem* — conjectured by Steinhaus, first proved by Sós, Surányi, and Świerczkowski [3, 4, 5, 6] — states that the gaps between successive returns of any irrational rotation take at most three distinct values, arranged quasi-periodically. Here they take two, $\{4, 5\}$. So the rigidity is not a special fact about τ : it is the universal rigidity of rotation return times. Once one grants a single dominant complex frequency, the strict two-value pattern is *forced*; nothing else was available.

2. THE SPECIFIC PATTERN IS THE CONTINUED FRACTION MADE VISIBLE

The average gap is $37/9 = 4.111\dots$, and as a continued fraction

$$\frac{37}{9} = [4; 9] = 4 + \frac{1}{9}.$$

That two-term expression *is* the observed law: the integer part 4 is the base gap; the partial quotient 9 is the period of the correction (one extra step — a 5 — every nine). The minima locations are, to this order, a Beatty sequence

$[k(37/9) + \beta]$ [7, 8]. So the “puzzle” of why an $8 \times 4 + 5$ block repeats with period 37 collapses to a two-term continued fraction. The Fourier peak at period 4.10 is the blurred analytic shadow of this crisp arithmetic object; the difference-table view resolves the object itself, which is why it looks far more rigid than a spectral peak has any right to.

3. WHERE THE INTERESTING NUMBER THEORY ACTUALLY LIVES

Two facts remain unexplained by the model.

First, that $H(t) = 1 + \sum_n \tau(p_n)t^n$ has a single, clean, well-isolated dominant complex zero at all. That one conjugate pair should govern everything is a genuine statement about the analytic structure of a *prime-indexed* τ series — an object with no modularity, no functional equation, and no Euler product. The dominant-singularity principle [12, 13] tells us what such a zero *would* control, but not why this series should have so clean a one. I know no reason.

Second, and more pointed: the value of the rotation number θ/π and its proximity to $9/37$. Is it *exactly* $9/37$ or merely very close? This is decidable, and the meaning hinges on it.

- If θ/π is *irrational* with continued fraction $[0; 4, 9, a_3, \dots]$ and a_3 large, then $9/37$ is an excellent rational approximant, the coding is Sturmian [1, 2], and near $n \sim 37a_3$ a *second-order defect* must appear — a block with seven or nine 4’s rather than eight, or two consecutive 5’s. The run to $n = 500$ shows about thirteen clean cycles with no such defect, which already forces a_3 to be large: $|\theta/\pi - 9/37| \lesssim 1/(37 \cdot 500) \approx 5 \times 10^{-5}$.
- If θ/π were *exactly* $9/37$, the pattern would be exactly period-37 forever. That would be astonishing: it would place the nearest zero of a prime-indexed τ series at a rational multiple of π , demanding an arithmetic explanation.

My honest bet is the first — irrational, $[0; 4, 9, a_3, \dots]$ with a_3 large — because there is no reason a transcendental-looking θ assembled from τ at primes should be a rational multiple of π , and every real number masquerades as a small-denominator rational when known to only four digits. I hold it loosely; the second case, if true, would be worth a paper.

4. HOW TO SETTLE IT

Pin θ down to eight or ten digits (rather than the four currently in hand), form θ/π , and read its continued fraction [8]. If it begins $[0; 4, 9, a_3, \dots]$, then a_3 *predicts the exact n at which the first second-order defect must occur* — a prediction tested by extending the run past $n = 500$ to that point. A defect at the predicted place proves the rotation-number account and shows θ/π irrational; no defect to all computable precision would be the surprising case. In the language of circle dynamics, this is just reading the rotation number and its denominators off the orbit [11].

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